

Things I've been building, lately

I've been having a lot of fun exploring projects lately. Below are just a handful of projects I've been messing with. Some are utilities. Some are learning projects. All of them have been incredibly fun and helped along with AI to various degrees.

// beagle

Dynamic, mostly-functional, compiles straight to machine code

No VM, no bytecode, no JIT warm-up. By far my longest-running, most complete project. All compilation and code generation use dependency-free custom backends for x86 and ARM. The goal is to be a language with fantastic live coding support allowing for humans or AI to build full applications without ever restarting a process.

● beagleexamples/echo_server.bg

```
1 use beagle.socket as socket
2
3 fn main() {
4   let listener = socket/listen("0.0.0.0", 8080)
5   println("Echo server listening on port 8080")
6
7   socket/on-connection(listener, fn(conn) {
8     loop {
9       let data = socket/read(conn, 4096) // fully async, no function coloring
10      if data == "" || data == null {
11        socket/close(conn)
12        break(null)
13      } else {
14        socket/write(conn, data)
15      }
16    }
17  })
18 }
```

FEATURE RICH LANGUAGE

01 Algebraic effects

Built-in algebraic effects make it easy to control and test IO

02 Live coding

Redefine functions, structs, and enums while your code is running

03 Async without function coloring

Delimited continuations give us the power to have transparent asynchronous

04 Compiled

No VM, no bytecode – direct machine code

// keep-running

Dtach, but with a friendlier CLI and a replay buffer

Run a long process, detach, let it keep going, reattach later from anywhere. Small utility, but I've found it incredibly useful.

● ● ● ~/keep-running

```
$ keep-running run -n build -- cargo build --release
→ session 'build' started · pid 84219
→ detach with ctrl-\\ ctrl-\\

$ keep-running ls
NAME  PID  STATUS  UPTIME  BUF
build 84219 running 14m    482 KB
tests 84482 running 3m     91 KB

$ keep-running attach build
→ replaying last 482 KB from buffer...
  Compiling porter v0.4.1 (/Users/_/code/porter)
  Compiling tokio v1.36.0
→ live
$
```

// bevy-whiteboard

The diagram on the whiteboard, but it actually runs

I've always wanted to be able to draw on a whiteboard and then truly simulate the system. This proof-of-concept does exactly that. Under the hood is a simulation driven by a rules-based language, enabling running simulations faster than real time or rewinding through time. It is quickly approaching something powerful and flexible enough to model real systems.

FLOW WHITEBOARD

TOOLS

SELECTCONNECT

EMITTERS

GENERATORCLIENTBACKOFFCLI

PROCESSORS

WORKERROUTERQUEUE

TERMINALS

SINKPROBE

DATA PALETTE

CLEAR

THEME

11201234567891011121314151617181920212223242526272829303132333435363738394041424344454647484950515253545556575859606162636465666768697071727374757677787980818283848586878889909192939495969798991001011021031041051061071081091101111121131141151161171181191201211221231241251261271281291301311321331341351361371381391401411421431441451461471481491501511521531541551561571581591601611621631641651661671681691701711721731741751761771781791801811821831841851861871881891901911921931941951961971981992002012022032042052062072082092102112122132142152162172182192202212222232242252262272282292302312322332342352362372382392402412422432442452462472482492502512522532542552562572582592602612622632642652662672682692702712722732742752762772782792802812822832842852862872882892902912922932942952962972982993003013023033043053063073083093103113123133143153163173183193203213223233243253263273283293303313323333343353363373383393403413423433443453463473483493503513523533543553563573583593603613623633643653663673683693703713723733743753763773783793803813823833843853863873883893903913923933943953963973983994004014024034044054064074084094104114124134144154164174184194204214224234244254264274284294304314324334344354364374384394404414424434444454464474484494504514524534544554564574584594604614624634644654664674684694704714724734744754764774784794804814824834844854864874884894904914924934944954964974984995005015025035045055065075085095105115125135145155165175185195205215225235245255265275285295305315325335345355365375385395405415425435445455465475485495505515525535545555565575585595605615625635645655665675685695705715725735745755765775785795805815825835845855865875885895905915925935945955965975985996006016026036046056066076086096106116126136146156166176186196206216226236246256266276286296306316326336346356366376386396406416426436446456466476486496506516526536546556566576586596606616626636646656666676686696706716726736746756766776786796806816826836846856866876886896906916926936946956966976986997007017027037047057067077087097107117127137147157167177187197207217227237247257267277287297307317327337347357367377387397407417427437447457467477487497507517527537547557567577587597607617627637647657667677687697707717727737747757767777787797807817827837847857867877887897907917927937947957967977987998008018028038048058068078088098108118128138148158168178188198208218228238248258268278288298308318328338348358368378388398408418428438448458468478488498508518528538548558568578588598608618628638648658668678688698708718728738748758768778788798808818828838848858868878888898908918928938948958968978988999009019029039049059069079089099109119129139149159169179189199209219229239249259269279289299309319329339349359369379389399409419429439449459469479489499509519529539549559569579589599609619629639649659669679689699709719729739749759769779789799809819829839849859869879889899909919929939949959969979989991000100110021003100410051006100710081009101010111012101310141015101610171018101910201021102210231024102510261027102810291030103110321033103410351036103710381039104010411042104310441045104610471048104910501051105210531054105510561057105810591060106110621063106410651066106710681069107010711072107310741075107610771078107910801081108210831084108510861087108810891090109110921093109410951096109710981099110011001110021100311004110051100611007110081100911010110111101211013110141101511016110171101811019110201102111022110231102411025110261102711028110291103011031110321103311034110351103611037110381103911040110411104211043110441104511046110471104811049110501105111052110531105411055110561105711058110591106011061110621106311064110651106611067110681106911070110711107211073110741107511076110771107811079110801108111082110831108411085110861108711088110891109011091110921109311094110951109611097110981109911100111001111002111003111004111005111006111007111008111009111010111011111012111013111014111015111016111017111018111019111020111021111022111023111024111025111026111027111028111029111030111031111032111033111034111035111036111037111038111039111040111041111042111043111044111045111046111047111048111049111050111051111052111053111054111055111056111057111058111059111060111061111062111063111064111065111066111067111068111069111070111071111072111073111074111075111076111077111078111079111080111081111082111083111084111085111086111087111088111089111090111091111092111093111094111095111096111097111098111099111100111100111110021111003111100411110051111006111100711110081111009111101011110111111012111101311110141111015111101611110171111018111101911110201111021111102211110231111024111102511110261111027111102811110291111030111103111110321111033111103411110351111036111103711110381111039111104011110411111042111104311110441111045111104611110471111048111104911110501111051111105211110531111054111105511110561111057111105811110591111060111106111110621111063111106411110651111066111106711110681111069111107011110711111072111107311110741111075111107611110771111078111107911110801111081111108211110831111084111108511110861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